

Feiyue Pan
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EDUCATION

School of Mathematical Sciences, Fudan University **Sep.2021-Jun.2025**
Bachelor of Science, major in Mathematics and Applied Mathematics **Shanghai,China**
- GPA: 3.56/ 4.00(90/100)
- Main courses: Mathematical Analysis/Advanced Algebra/University Physics/Probability Theory/Statistical Inference/Time Series Analysis/Multivariate Analysis/Categorical Data Analysis
- Awards: First Prize in National Undergraduate Mathematics Competition(twice in 2022 and 2023)/ First Prize in Shanghai Mathematics Competition(twice in 2022 and 2023) / Undergraduate Scholarship of Fudan University / Fudan University "Hehechang" Scholarship/ Major Scholarship of Fudan University

INTERNSHIP EXPERIENCE

Galaxy Derivatives Capital Management Ltd. **Jan.2024-Feb.2024**
(Derivatives trading)
Position: Assistant Researcher, Market Making Business Department
- I was responsible for organizing, researching, and analyzing post-market and historical data using Python. During my internship, I primarily focused on studying the long-term and intraweek variations in the price ratio of gold and silver futures (gold-silver ratio). I provided comprehensive analysis results and proposed three trading strategies, all of which demonstrated backtested and forecasted annual returns of around 20%. Additionally, I investigated the impact of large-scale trading on the prices of commodities such as palm oil, ethylene glycol, and styrene, preliminary findings presented.

RESEARCH EXPERIENCE

Fudan University **Jul.2024-Aug.2024**
School of Mathematical Sciences, Fudan University
-Under the supervision of Liu Yanchao, a second-year doctoral student at the School of Mathematical Sciences, Fudan University, I am assisting in a research project related to airline network optimization. The primary objective of this project is to analyze and optimize the topological structure and layout efficiency of the route network of Chinese airlines. By applying statistical analysis and operations research methods, we aim to enhance the operational efficiency of the airline network and reduce operational costs.

Fudan University **Aug.2024-May.2025**
Shanghai Center for Mathematical Sciences
- Under the guidance of Dr. Wang Tiandong, Assistant Professor at the Shanghai Center for Mathematical Sciences, Fudan University, I am conducting independent research as an undergraduate. This study focuses on the optimization of the Weighted PageRank algorithm under specific requirements. The PageRank algorithm determines the importance ranking of nodes based on the connections between them. However, certain nodes, resembling marketing pages, may undesirably attain unjustly high rankings. To address this issue, we propose two original correction methods based on k-core decomposition. Currently, the majority of the research has been completed.

RESEARCH EXPERIENCE

Fudan University

May.2025-

Shanghai Institute for Mathematics and Interdisciplinary Sciences

-Under the supervision of Professor Zhigang Yao at the National University of Singapore (NUS), I advanced an independent summer research project. The project aims to leverage deep learning and manifold fitting to mine UK Biobank disease data, investigate latent patterns between NMR features and patients' diseases, and, on this basis, develop an interactive analytical platform. Furthermore, we aim to extend the utility of manifold fitting methods through this project, particularly for stratifying disease data and matching new data to existing cohorts.

ACADEMIC CONFERENCE PARTICIPATING

Fudan University

Jul.2024

Shanghai Center for Mathematical Sciences

-I participated in the International Conference on Extremes, Statistics, and Quantitative Risk Management held at the Shanghai Center for Mathematical Sciences, and assisted in completing various tasks related to the organization and execution of the conference.